TokioMarine Specialty Insurance - Underwriting Acceleration POC

4-Week Sprint Outline — Definite vs. Aspirational

Business Value Summary

The Problem Today

TokioMarine Specialty receives ~3,000 submissions annually and quotes roughly 1,500 of them. The current submission-to-quote cycle takes ~20 days — driven by manual email triage, hand-keyed rating workbook entry, manual data cleaning for CAT models, fragmented claims lookups, and underwriter time spent hunting for comparable risks and institutional knowledge across disconnected systems. The remaining ~1,500 submissions are either declined or never fully evaluated, representing unrealized capacity.

The Opportunity

Compressing the quote cycle from 20 days to 1-2 days unlocks compounding value across four dimensions:

Value Lever	Assumption	Estimated Impact
Quote more submissions	At 1-2 day turnaround, UW capacity increases ~10x on cycle time alone. Conservatively assume TokioMarine can evaluate 50-100% more submissions (4,500+) by eliminating manual bottlenecks in triage, data entry, and comp research.	+1,500+ additional submissions evaluated per year
Win more quotes	Brokers place business with carriers who respond fast. Industry data shows first-to-quote wins disproportionately. Assume a 5-15% improvement in bind rate on quoted submissions from speed-to-market alone.	+75-225 additional bound policies per year

Write better policies	Agent-assisted synthesis surfaces claims comps, similar risk history, and UW notes that are hard to find manually today. Better-informed pricing → fewer underpriced risks, more profitable book. Assume 1-3% improvement in loss ratio from better risk selection and pricing accuracy.	Direct improvement to combined ratio
Unlock net new intelligence	Comparable loss reports (modeled from claims data on similar risk profiles) don't exist today. This is a new analytical asset that improves pricing confidence on new business where no account history exists.	New capability — currently a blind spot

Compounding Effects

- **Faster triage = capacity to expand the funnel.** If TokioMarine can process submissions in hours instead of weeks, they can accept flow from more brokers or enter adjacent segments without adding headcount.
- **Faster quotes = broker preference.** Consistent fast response builds broker relationships and repeat flow. TokioMarine becomes the first call, not the backup.
- **Better data at point of decision = fewer NTUs and better terms.** Underwriters armed with comps, claims, and institutional knowledge quote with more conviction. Fewer quotes get walked back or not taken up.
- **UW time recaptured.** The highest-value activity an underwriter performs is judgement — not data entry, not searching for comps, not cleaning files for CAT models. Every hour of manual work automated is an hour returned to underwriting judgement.

Conservative Scenario

Metric	Current State	With Acceleration
Submissions received	~3,000/yr	4,500+/yr (expanded capacity)
Submissions evaluated	~1,500/yr	3,000-4,000/yr
Quote turnaround	~20 days	1-2 days
Bind rate improvement	Baseline	+5-15% from speed-to-market
Loss ratio impact	Baseline	-1-3% from better risk selection
Net new assets	None	Comparable loss reports for all quotes

Quantified Value Framework ($250M GWP Base)

Current economics (estimated):

- ~3,000 submissions/yr → ~1,500 quoted → ~750 bound (illustrative 50% bind rate)
- ~$250M GWP → ~$167K average premium per bound policy
- ~20-day quote cycle limits throughput and broker preference

Value Lever 1: Top-Line Growth — $12M-$50M Incremental GWP

Driver	Math	Range
Evaluate more submissions	2x throughput capacity (1,500 → 3,000 evaluated). Even if decline rate holds, doubles the quoted volume.	+750 additional quotes
Improve bind rate from speed	First-to-quote advantage: 5-15% bind rate improvement on all quoted submissions. At 2,250 quotes × 5-15% incremental bind rate × $167K avg premium.	**$19M-$56M**
Conservative (blended)	Assume only partial throughput gain and modest bind improvement.	**$12M-$25M incremental GWP**

At a 10%+ target return on allocated capital, every dollar of well-priced GWP is highly accretive.

Value Lever 2: Underwriting Profit — $2.5M-$7.5M/yr

Driver	Math	Range
Loss ratio improvement	Better risk selection via claims comps, similar risk analysis, and UW notes mining. Every 1pt of loss ratio on $250M = $2.5M.	**$2.5M-$7.5M/yr** (1-3pts)
Avoided underpricing	Agent surfaces adverse claims trends and comparable losses that UWs might miss manually. Reduces frequency of underpriced risks entering the book.	Embedded in loss ratio improvement

NTU reduction	Faster, better-informed quotes → fewer quotes walked back or not taken up. Each NTU is wasted UW capacity. Even a 5% NTU reduction recovers meaningful bound premium.	Incremental to bind rate improvement above

On a $250M book targeting 10%+ risk-free rate on allocated capital, 1-3 points of loss ratio is the difference between a good year and a great year.

Value Lever 3: Capacity & Efficiency — 7-10 FTE Equivalent

Driver	Math	Range
UW time recaptured	Assume 15-20 UWs spend ~50% of time on manual tasks (data entry, file cleaning, comp searching, workbook population). Automation reclaims 50%+ of that time.	**7-10 FTE equivalent capacity**
Headcount avoidance	To grow from $250M to $300M+ organically would typically require proportional UW hiring. Agent-assisted workflow absorbs volume growth without linear headcount growth.	Avoid 3-5 hires as book grows
Scalability	Current process is linearly bound to human throughput. Agent pipeline scales sub-linearly — processing cost per submission drops as volume increases.	Structural operating leverage

This isn't about replacing underwriters — it's about giving the best underwriters in the market leverage to apply their judgement to 2-3x the opportunity set.

Total Addressable Value (Annual, Steady State)

Category	Conservative	Upside
Incremental GWP	$12M	$50M+
Underwriting profit improvement	$2.5M	$7.5M
Capacity equivalent (FTEs)	7	10+
Combined financial impact	**$15M-$20M/yr**	**$50M+/yr**

These estimates assume the system reaches production maturity. The POC validates the feasibility and quality of each value lever so TokioMarine can invest with confidence.

What This POC Proves

This 4-week POC demonstrates the end-to-end pipeline on a representative sample of real submissions. It validates that AI-driven submission parsing, guideline checks, rating input mapping, similar risk retrieval, claims intelligence, and UW synthesis can produce outputs that underwriters trust and would use — at a fraction of the current cycle time. Success here justifies production investment to capture the full value above.

POC Philosophy: Prove the art of the possible on a narrow slice, not perfection across all cases. Cut corners on coverage breadth, solve for depth on the happy path. Production solves for scale, edge cases, and full automation.

Week 1: Data Foundation & Submission Intelligence

Definite Deliverables

#	Deliverable	Details
D1	All raw data landed in Snowflake	Bulk load emails, promised lines, rating data (ERS), rating data sheets, claims (ERS), UW guidelines, and authority letters. No transformation yet — just land it.
D2	Entity resolution: "good enough" linkage	Use fuzzy matching on insured name + date + broker to link submissions → promised lines → rating data → claims. Accept ~70-80% match rate. Don't boil the ocean — manual mapping for demo cases is fine.
D3	UW notes corpus built and searchable	Extract notes/commentary from rating data sheets and promised lines (decline reasons, UW comments). Load into Cortex Search Service. This is the highest-signal data — prioritize it.
D4	Email submission parser (5-10 representative examples)	Parse a curated set of real submissions end-to-end: extract broker, insured, coverage type, key terms, attachments. Use Cortex AI Functions (PARSE_DOCUMENT) for attachments. Don't try to handle all formats — pick the most common patterns.

D5	Workbook input/output mapping documented	Catalog the template rating workbook: what goes in, what comes out, which sheets matter. We need this to know what the submission parser must extract.

Aspirational Deliverables

#	Deliverable	Details	Unlock Condition
A1	Submission completeness scoring	Auto-score whether a submission has enough info to rate. Flag gaps, generate a broker follow-up request.	D4 works reliably on the curated set
A2	Broker scorecard from historical data	Profile brokers by historical submission quality, completeness rates, win rates.	D2 entity resolution covers enough data
A3	Email classification (initial vs. mid-journey)	Classify the 15k emails into initial submissions vs. follow-ups/mid-journey. Needed for production but not for the curated demo path.	D1 + D4 done early

POC Shortcuts (Solve in Production)

- **Skip full entity resolution** — for the POC, manually verify linkage on 20-50 demo cases rather than building a production-grade matching pipeline.
- **Curate the submission set** — hand-pick 5-10 submissions that represent common patterns (Lead, Follow, different LOBs). Don't try to handle the full 15k.
- **Attachment parsing: top 3 formats only** — pick the most common attachment types (likely SOV Excel, PDF application, loss run PDF). Ignore edge cases.

Week 2: Guideline Checks & Rating Pipeline

Definite Deliverables

#	Deliverable	Details
D6	Deterministic guideline check engine	Encode the core UW guidelines as pass/fail rules (appetite, limits, territories, excluded classes). Run against parsed submission fields. This is deterministic today — replicate it as a rule set in SQL/Python.
D7	Authority letter check	Check submission parameters against authority limits. Flag referral triggers. Simpler than guidelines — just threshold checks.
D8	Rating workbook: input mapping pipeline	Automated mapping from parsed submission fields → workbook input fields. This is the hardest translation layer. For POC: build for the curated set, accept manual overrides.
D9	Rating workbook execution (or simulation)	**POC shortcut option:** Rather than fully replicating the 12-15 sheet workbook programmatically, consider: (a) calling the Excel workbook via a headless Excel engine, OR (b) using historical rating data sheets as "ground truth" and demonstrating the agent can produce inputs that would generate the known outputs. Full replication is a production effort.

Aspirational Deliverables

#	Deliverable	Details	Unlock Condition
A4	Guidelines vs. past decisions comparison	For a submission, surface historical cases where similar risks were approved/declined and what the UW rationale was. Shows where the book deviated from guidelines.	D6 + D3 (notes corpus) working
A5	CAT model input preparation	Automate the manual data cleaning step that feeds RMS/CAT models. This is a huge time sink today — even partial automation is valuable.	D8 input mapping is solid
A6	Full programmatic rating engine	Replicate the workbook logic in code. Only attempt if D9 shortcut proves insufficient for the demo.	D9 shortcut validated + team bandwidth

POC Shortcuts (Solve in Production)

- **Rating workbook: don't replicate, validate** — Instead of rebuilding the 12-15 sheet Excel model in code, use historical input/output pairs to show the agent can correctly extract and map inputs. Demo the translation, not the calculation. Actual workbook execution can use Optalytics or a headless Excel approach in production.
- **CAT models: use historical outputs** — Stub CAT model results with historical data for the demo cases. Prove the pipeline can consume CAT outputs, not that it can call RMS.
- **Guideline encoding: start with the top 10 rules** — Don't encode every guideline. Pick the 10 most common decline reasons from promised lines data and build checks for those.

Week 3: Intelligence Layer & Synthesis

Definite Deliverables

#	Deliverable	Details
D10	Similar risk search service	Cortex Search over historical submissions, ratings, and notes. Given a new submission, retrieve the most comparable past risks by industry, geography, coverage type, TIV range, occupancy.
D11	Past ratings & decisions for similar risks	For retrieved similar risks: surface technical pricing, rate changes, terms quoted, win/loss, and UW commentary. This gives underwriters market context and internal precedent.
D12	Claims analysis — account history (renewals)	For renewal submissions: pull complete claims history, summarize loss experience, flag large losses and trends. This is straightforward if entity resolution worked.
D13	Claims analysis — comparable risk profiles (new biz)	For new business: find claims from similar risk profiles and model expected loss. **This is the net new asset** — the loss report they don't have today. Even a rough version is high value.
D14	UW synthesis agent v1	Cortex Agent that takes all outputs (parsed submission, guideline checks, rating inputs/outputs, similar risks, claims) and produces a structured underwriter decision support package. First version — get the structure right, tune quality in Week 4.

Aspirational Deliverables

#	Deliverable	Details	Unlock Condition
A7	Quote/no-quote recommendation with confidence	Agent generates an explicit recommend/decline with confidence score and evidence trail. Goes beyond synthesis into actual decisioning support.	D14 synthesis quality is strong
A8	Draft quote terms	For submissions the agent recommends quoting: generate draft terms, pricing summary, and conditions.	A7 + D9/D11 data is reliable
A9	Notes-mined "institutional knowledge" layer	Beyond similar risk search — extract and synthesize recurring UW themes, market commentary, and pricing philosophy from the notes corpus. Surface as contextual intelligence.	D3 notes corpus is rich enough

POC Shortcuts (Solve in Production)

- **Comparable claims: simple similarity, not actuarial modeling** — Use attribute-based matching (industry, geography, size) to find comps, then summarize their loss experience. Don't build a loss model — just show relevant claims data. Production can layer in actuarial rigor.
- **Synthesis output: structured markdown, not a polished UI** — The agent output can be a well-structured markdown/text report. Don't spend time on pixel-perfect formatting. Streamlit wrapper is nice-to-have.
- **Similar risk retrieval: semantic search is the 80/20** — Don't build a complex multi-dimensional scoring model. Cortex Search with good embeddings on submission descriptions + notes gets you most of the way there.

Week 4: Validation, Demo & Readout

Definite Deliverables

#	Deliverable	Details
D15	End-to-end demo on 5-10 curated submissions	Full pipeline walkthrough: email in → parsed submission → guideline check → rating inputs → similar risks + claims → UW synthesis out. Show real data, real outputs.
D16	Backtest against known outcomes	Run the pipeline against historical submissions where we know the actual UW decision. Show: did the agent flag the same guideline issues? Did it find the right comps? Did the synthesis align with the UW's actual decision?
D17	UW feedback session	Walk TokioMarine underwriters through agent outputs. Collect structured feedback: accuracy, completeness, usefulness, trust. This is the most important validation — does the UW find this useful?
D18	POC findings report & production roadmap	Document: what worked, what didn't, accuracy metrics, UW feedback, identified gaps, and a clear path from POC to production with effort estimates.

Aspirational Deliverables

#	Deliverable	Details	Unlock Condition
A10	Streamlit UI for the demo	Interactive app where you paste/upload a submission email and watch the agent pipeline process it step-by-step with visible outputs at each stage.	D15 pipeline is stable and fast enough
A11	Broader backtest (50+ submissions)	Expand beyond the curated set to test robustness. Surface failure modes and edge cases systematically.	D16 on curated set passes cleanly
A12	Metrics dashboard	Automated measurement: processing time, extraction accuracy, guideline check	A11 broader backtest running

		accuracy, retrieval relevance, synthesis quality scores.	
A13	Production architecture & cost estimate	Detailed technical design for production: data pipelines, agent architecture, monitoring, retraining, and estimated Snowflake compute costs.	D18 findings are clear enough to extrapolate

POC Shortcuts (Solve in Production)

- **Backtest scope: quality over quantity** — 5-10 well-analyzed cases with known outcomes beats 100 superficially checked ones. Show depth of analysis on the demo cases.
- **UW feedback: live session, not a survey** — Sit with underwriters, walk through outputs, take notes. Don't build a formal feedback collection system.
- **Demo over dashboard** — A live walkthrough with a notebook or simple Streamlit app beats a polished metrics dashboard. Invest time in making the agent outputs impressive, not the wrapper.

Summary: Definite vs. Aspirational at a Glance

Definite (Must Ship)

Wk	What You'll Show
1	"We can ingest a submission email, parse it, extract the key fields, and search across your entire UW notes history."
2	"We can check it against your guidelines, flag referrals, and map the extracted data to your rating workbook inputs."
3	"We can find similar risks, pull claims data, and produce a complete UW synthesis report that combines everything."
4	"Here's the end-to-end pipeline on real submissions. Here's how it compares to what your UWs actually decided. Here's what they think of it."

Aspirational (Ship if Fast)

Wk	What You'll Also Show
1	Completeness scoring, broker scorecards, email classification
2	Guidelines vs. past decisions analysis, CAT input prep, programmatic rating engine
3	Quote/no-quote recommendations, draft terms, institutional knowledge extraction
4	Interactive Streamlit UI, broad backtest, metrics dashboard, production architecture

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Critical Path & Biggest Risks

Risk	Impact	Mitigation
Rating workbook complexity blocks Week 2	Can't show rating pipeline	Use the "validate inputs, don't replicate engine" shortcut. Show we can produce correct inputs; defer engine replication.
Entity resolution is poor across datasets	Claims and similar risk features are weak	Manually verify linkage for demo cases. Be transparent about match rates. Production invests in proper MDM.
Submission attachments are too varied to parse	Intake agent fails on real data	Curate demo set to common formats. Show it works on 70% of cases, document the long tail for production.
UW notes are sparse or low-signal	Similar risk and synthesis quality suffers	Discover this in Week 1. If true, pivot emphasis to structured data (rating sheets, claims) over unstructured notes.
TokioMarine data delivery is late	Everything shifts right	Get data access confirmed before Day 1. Have a fallback plan using synthetic/sample data to build pipeline skeleton.

Technology Stack (Snowflake Platform)

Component	Technology

Data Storage & Processing	Snowflake (tables, stages)
Document Intelligence	Cortex AI Functions (PARSE_DOCUMENT, AI_EXTRACT)
Semantic Search (notes, similar risks)	Cortex Search Service
Agent Orchestration	Cortex Agent
UW Synthesis & Reasoning	Cortex AI (LLM completions)
Rating Model Execution	Historical validation approach for POC; Optalytics / headless Excel for production
CAT Model Integration	Stubbed with historical outputs for POC; API integration for production
Presentation Layer	Notebook / simple Streamlit for POC
Guidelines Rule Engine	SQL / Python stored procedures

Success Criteria for POC

Metric	POC Target	Production Target
End-to-end processing time	< 30 min per submission	< 10 min
Submissions demonstrated	5-10 curated	All incoming
Submission parsing accuracy	High on curated set (>90%)	>90% across all formats
Guideline check accuracy	100% on encoded rules	100% on all rules
Similar risk retrieval relevance	UW-validated as relevant	Measured precision/recall
Claims comp quality	UW-validated as useful	Actuarially sound

UW satisfaction with synthesis	"I would use this"	>4/5 formal rating
Comparable loss report (net new)	Demonstrated on sample	Standard deliverable for all quotes